Digital Chip Design Program

RTL Design

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Task#4

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Structural modeling & testbench

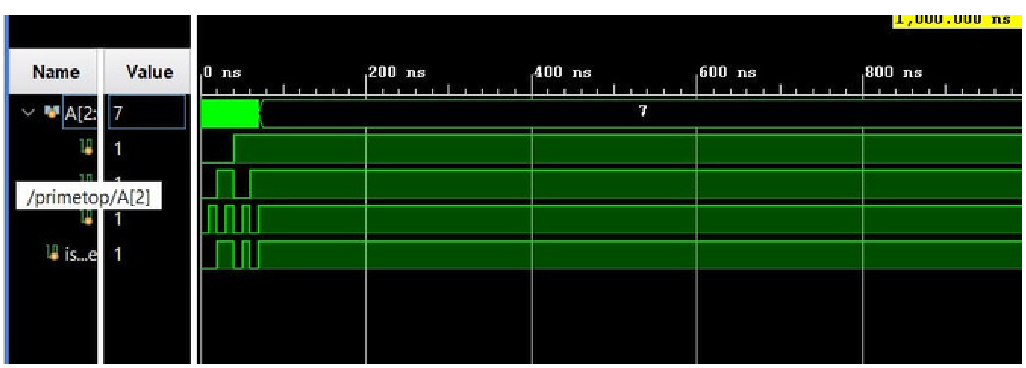
Task #4.1

```
module MyTop (a ,b, c , n1 ,y);
input [3:0] a,b;
input logic c ;
output logic n1 , y;
logic n2 ,n3 ,n4;

or2 inst1(.a(n3),
.b(n4),
.y(n1));

and2 inst2(.a(c),
.b(c),
.y(n2));

not1 inst3(.a(n2),
.y(y));

comparator inst4(.a(a),
.b(b),
.AeqB(),
.AltB(n4),
.AgtB(n3));

endmodule
```

```
module or2 (a ,b ,y);
input logic a ,b;
output logic y;
assign y = a | b;
endmodule

module and2 (a ,b ,y);
input logic a ,b;
output logic y;
assign y = a & b;
endmodule

module not1 (a ,y);
input logic a ;
output logic y;
assign y = ~a ;
endmodule

module comparator(AeqB, AltB, AgtB, a, b);
input [3:0] a,b;
output AeqB, AltB, AgtB;
assign AeqB = (a == b)? 1 : 0;
assign AltB = (a < b)? 1 : 0;
assign AgtB = (a > b)? 1 : 0;
endmodule
```

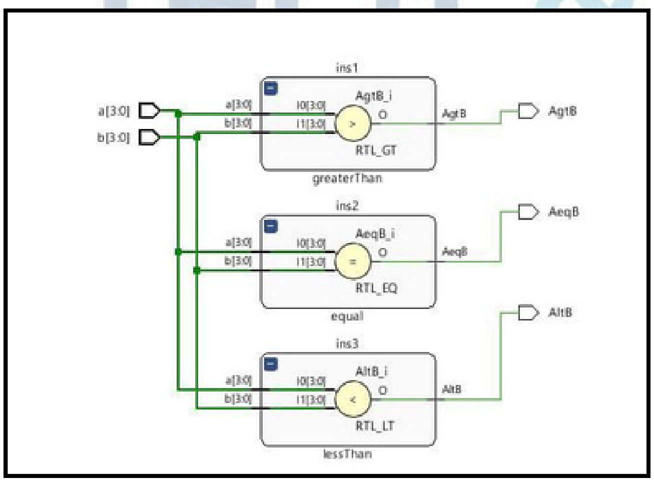
Task #4.2

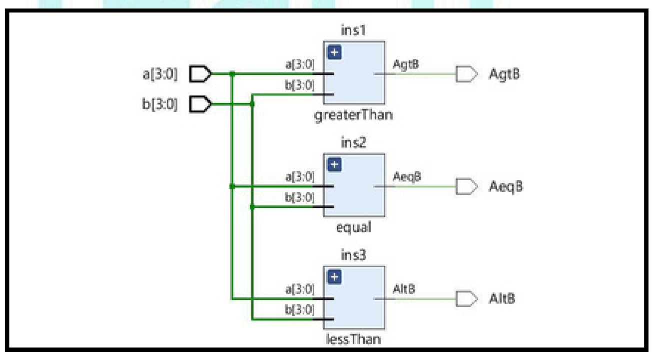
```
module primetop();

logic [2:0] A;
logic isPrime;

prime dut(
   .A(A),
   .isPrime(isPrime)
);

initial begin
A = 3'b000; #10;
A = 3'b001; #10;
A = 3'b010; #10;
A = 3'b011; #10;
A = 3'b100; #10;
A = 3'b101; #10;
A = 3'b110; #10;
A = 3'b111; #10;

$finish;
end
endmodule
```

Task #4.3

```
module task_4(a , b , AgtB ,AeqB ,AltB );

input [3:0]a , b ;
output AgtB ,AeqB ,AltB ;

// greaterThan
greaterThan ins1( .a(a),
            .b(b),
            .AgtB(AgtB)
);

// equal
equal ins2( .a(a),
            .b(b),
            .AeqB(AeqB)
);

// lessThan
lessThan ins3( .a(a),
            .b(b),
            .AltB(AltB)
);

endmodule
```



Task #4.3

```
`timescale 1ns / 1ps
module Testbench();
  logic [3:0] a,b;
  logic AgtB ,AeqB ,AltB ;

  task_4 dut (
          .a(a),
          .b(b),
          .AgtB(AgtB),
          .AeqB(AeqB),
          .AltB(AltB)
  );
  initial begin
  a=0; b=0;
  for(int i=0; i<5'b10000; i++)  begin
     for(int j=0; j<5'b10000; j++)  begin
        #10 a=i; #10 b=j;
      end
     end
     #10 $finish;
 end
endmodule
```


Work distribution

Section	ROAA	RAND	LANA	LARA	RAWAN	RENAD
Report			✓	✓		
Task 3.1			✓	✓		
Task 3.2		✓				
Task 3.3					✓	✓